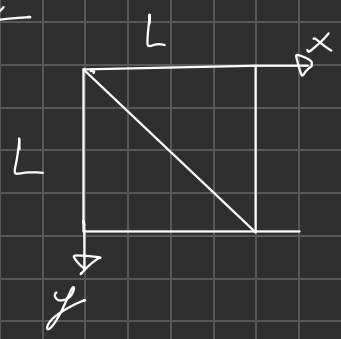
Task 1

1.1)



a) $y = 0$, $x \in [0, L] \Rightarrow \rho \in [0, x] = [0, L]$

b) $y = 0$, $x = 0 \Rightarrow \rho = 0$

c) $\rho = x \cos\theta + y \sin\theta$

$\rho = x \cos\frac{\pi}{4} + y \sin\frac{\pi}{4}$

$x \wedge y = 0 \Rightarrow \rho = 0$

$x \wedge y = L \Rightarrow \rho = L \cos\frac{\pi}{4} + y \sin\frac{\pi}{4}$

$\rho = \sqrt{2}L$

$\rho \in [0, \sqrt{2}L]$

d) $\rho = 0$

1.2) a) Yes, since we use $\nabla I(x)$ and constants disappear.

$$\frac{d}{dx} I = \nabla I$$

$$\frac{d}{dx}(I + c) = \nabla I$$

b) If ω is radially symmetric, angle doesn't matter. This means that no matter the $\Delta\theta$, it won't change, since the only thing that will change is when changing r.

1.3) I would assume that it does not affect which camera our detected, since the acceptance thresholds is specified relatively and not according to the standard pixel values. This will only work if no values gets clipped

2.3) $$\frac{row}{N_p}\underline{(P_{max} - P_{min})} + P_{min} = P$$

P gtheu

T3